

Statistics Method II Assignment

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February 11, 2025

Question one:

Codes:

```
#Data Importation
mydata <- read.csv("~/R for Statistics and Data Science (STAT-5521G)/Stat_meth.txt", sep="")

#Linear Regression Model
lm_mdel<-lm(Minutes~Copiers, data = mydata)

#Summary of the LM result
summary(lm_mdel) -> summary_mdel
summary_mdel

##
## Call:
## lm(formula = Minutes ~ Copiers, data = mydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## Copiers       15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

(a.) What is the dependent variable in this case;

The dependent variable is: **total number of minutes.**

(b.) What is the independent variable in this case;

The independent variable is: **number of copiers serviced.**

(c.) Show what is the theoretical simple linear regression model;

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where:

- Y = Total service time (minutes) (**dependent variable**),
- X = Number of copiers serviced (**independent variable**),
- β_0 = Intercept (baseline service time when no copiers are serviced),
- β_1 = Slope (additional time required per copier).

(d.) Can you provide a least squares estimator for the parameter value β_1 ;

$$\beta_1 = 15.035248$$

(e.) Can you provide a least squares estimator for the parameter value;

$$\beta_0 = -0.5801567$$

(f.) Show what is the fitted linear regression model;

Total service time (minutes) = $-0.5801567 + 15.035248$ (Number of copiers serviced)

(g.) Can you provide an interpretation for the slope β_1 ;

For one unit increase in number of copiers serviced, will lead to increase in the total serviced time on average by 15.035248.

(h.) Can you provide an interpretation for the intercept β_0 ;

On average the total serviced time is -0.5801567 whenever the number of copiers serviced is 0.